# FDTD Equations Reference

From `fdtd_brain_simulation_engine.py`

This document lists and labels all physical and mathematical equations implemented in the 3D FDTD brain simulation engine. Equation numbers match the order of use in the code (time step → materials → field updates → SAR → thermal).

## 1 Time Step (Courant Condition)

The time step is set from the Courant stability condition for a uniform grid ($\Delta x = \Delta y = \Delta z = dx$). Implemented at engine setup (e.g. `fdtd_brain_simulation_engine.py` around lines 2614–2621, and in `_run_minimal_fdtd_benchmark` 185–188).

$$\Delta t_{\text{courant}} = \frac{1}{c_0\sqrt{1/(\Delta x)^2 + 1/(\Delta y)^2 + 1/(\Delta z)^2}} \tag{1}$$

with $c_0 = 2.99792458 \times 10^8\,\text{m/s}$ (speed of light in vacuum).

$$\Delta t = F \cdot \Delta t_{\text{courant}} \tag{2}$$

where $F$ is the Courant factor (default 0.99), set by `-courant-factor`.

## 2 Material Coefficients (ADE for Lossy Media)

Lossy, dispersive-free materials use an auxiliary differential equation (ADE) in the engine. Relative permittivity $\varepsilon_r$, conductivity $\sigma$, and mass density $\rho$ are assigned per tissue from a lookup table. The effective coefficients used in the E-field update are computed once per voxel (engine lines 2683–2691):

$$c = \frac{1}{\varepsilon_r + \sigma\Delta t/\varepsilon_0}, \qquad \text{cond} = \frac{\sigma\Delta t}{\varepsilon_0} \tag{3}$$

with $\varepsilon_0 = 8.854 \times 10^{-12}\,\text{F/m}$. The arrays `eps_x`, `eps_y`, `eps_z` store $c$; `conductivity_x/y/z` store `cond`.

## 3 FDTD Field Updates (Yee Scheme + PML)

### 3.1 D-field update (curl of H)

Displacement field $\mathbf{D}$ is updated from $\nabla \times \mathbf{H}$ on a staggered grid. PML is applied via coefficients $g_{i1}, g_{i2}, g_{i3}$ (and similarly $g_{j*}$, $g_{k*}$). Implemented in `calculate_dx_field`, `calculate_dy_field`, `calculate_dz_field` (lines 384–462).

Example for $D_x$ (one component): with

$$(\nabla \times \mathbf{H})_x = \frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z},$$

the discrete curl at $(i, j, k)$ is

$$\text{curl_h} = H_z|_{i,j,k} - H_z|_{i,j-1,k} - H_y|_{i,j,k} + H_y|_{i,j,k-1}. \tag{4}$$

The update uses PML auxiliary variable $iD_x$ and coefficients $g_{j3}, g_{k3}, g_{j2}, g_{k2}, g_{i1}$:

$$iD_x \leftarrow iD_x + \text{curl_h}; \qquad D_x \leftarrow g_{j3}g_{k3}D_x + g_{j2}g_{k2}\left(\tfrac{1}{2}\,\text{curl_h} + g_{i1}\,iD_x\right). \tag{5}$$

$D_y$ and $D_z$ follow from the other components of $\nabla \times \mathbf{H}$ with the corresponding PML indices.

### 3.2 E-field from D-field (ADE)

Electric field $\mathbf{E}$ is obtained from $\mathbf{D}$ and the integrated conduction current $\mathbf{I}$ (engine: `calculate_e_fields`, lines 484–514):

$$E_\alpha = c_\alpha\,(D_\alpha - I_\alpha), \qquad I_\alpha \leftarrow I_\alpha + \text{cond}_\alpha\,E_\alpha \tag{6}$$

for $\alpha \in \{x, y, z\}$, with $c$ and cond from (3).

### 3.3 H-field update (curl of E)

Magnetic field $\mathbf{H}$ is updated from $\nabla\times\mathbf{E}$ with PML (e.g. `calculate_hx_field`, `calculate_hy_field`, `calculate_hz_field`, lines 544–619). Example for $H_x$:

$$\text{curl_e} = E_y|_{i,j,k+1} - E_y|_{i,j,k} - E_z|_{i,j+1,k} + E_z|_{i,j,k}; \tag{7}$$

$$iH_x \leftarrow iH_x + \text{curl_e}; \qquad H_x \leftarrow f_{j3}f_{k3}H_x + f_{j2}f_{k2}\,\tfrac{1}{2}\,(\text{curl_e} + f_{i1}\,iH_x). \tag{8}$$

$f_*$ are the PML coefficients for the H-update (lines 299–357: cubic grading $x_n = 0.33\,((n_{\text{PML}} - n)/n_{\text{PML}})^3$).

## 4 SAR (Specific Absorption Rate)

### 4.1 Instantaneous SAR

Used per time step when accumulating E-field squared (`sar_instant`, `compute_sar_instantaneous`; see lines 762–784):

$$\text{SAR} = \frac{\sigma\,|\mathbf{E}|^2}{2\rho}, \qquad |\mathbf{E}|^2 = E_x^2 + E_y^2 + E_z^2. \tag{9}$$

Units: W/kg. $\sigma$ is averaged over $x, y, z$ components when they differ (isotropic assumption).

### 4.2 SAR from time-averaged (RMS) E-field

After accumulating $\sum E_x^2$, $\sum E_y^2$, $\sum E_z^2$ over $N_{\text{samples}}$ steps (`compute_sar`, lines 803–826):

$$|\mathbf{E}_{\text{rms}}|^2 = \frac{1}{N_{\text{samples}}}\left(\sum E_x^2 + \sum E_y^2 + \sum E_z^2\right); \qquad \text{SAR} = \frac{\sigma\,|\mathbf{E}_{\text{rms}}|^2}{2\rho}. \tag{10}$$

### 4.3 SAR from complex phasor (antenna optimization)

In the optimization path, SAR is computed from the synthesized complex field $\mathbf{E}_{\text{total}}$ (`compute_sar_from_complex_f` lines 1077–1085):

$$\text{SAR} = \frac{\sigma\,|\mathbf{E}_{\text{total}}|^2}{2\rho}, \qquad |\mathbf{E}_{\text{total}}|^2 = \left|\text{Re}(\mathbf{E}) + \mathrm{i}\,\text{Im}(\mathbf{E})\right|^2. \tag{11}$$

# 5 Thermal Solver (Pennes Steady-State)

## 5.1 Heat source from SAR

Volumetric heat source (W/m$^3$) from SAR deposition (engine: $Q_\text{heat} = \text{SAR} \cdot \rho$, line 4358; see also comment at 4349):

$$Q = \text{SAR} \cdot \rho. \tag{12}$$

## 5.2 Steady-state bioheat (no perfusion)

Simplified Pennes equation with constant thermal conductivity $k$ and no perfusion (`solve_steady_bioheat_3d`, docstring and implementation around 831–865):

$$\nabla \cdot (k \nabla T) + Q = 0. \tag{13}$$

Dirichlet boundary condition: $T = T_{\text{boundary}}$ (default 37 °C) on the domain boundary.

## 5.3 Gauss-Seidel update

Central-difference Laplacian on a uniform grid with spacing $dx$; Gauss-Seidel iteration over interior voxels with $k > 0$ (lines 844–861):

$$T_{i,j,k} = \frac{1}{6}\left(T_{i+1,j,k} + T_{i-1,j,k} + T_{i,j+1,k} + T_{i,j-1,k} + T_{i,j,k+1} + T_{i,j,k-1} + \frac{Q_{i,j,k}\,(\Delta x)^2}{k_{i,j,k}}\right). \tag{14}$$

Iteration stops when $\max|T - T_{\text{old}}| < \text{tol}$ (default $10^{-6}$) or after a maximum number of iterations.

# 6 Antenna Optimization (J-ratio)

## 6.1 J-ratio objective

Definition of the optimization figure of merit (`compute_j_ratio`, lines 1089–1098):

$$J = \frac{\overline{\text{SAR}}_{\text{tumor}}}{\overline{\text{SAR}}_{\text{healthy}}}. \tag{15}$$

Overlines denote spatial mean over the respective tissue masks.

## 6.2 Penalized objective (optional)

When a penalty weight $w$ is used to limit healthy-tissue hotspots (`compute_robust_objective`, 1102–1122):

$$J_{\text{eff}} = J - w\,\frac{P_{95}(\text{SAR}_{\text{healthy}})}{\overline{\text{SAR}}_{\text{tumor}}}. \tag{16}$$

$P_{95}$ denotes the 95th percentile of SAR in the healthy-tissue region.

# 7 Source and Ramp

## 7.1 CW soft source with Gaussian ramp

For continuous-wave or sinusoidal sources, the soft source adds to $D_z$ at the source gap (e.g. lines 1005–1008, 3622–3625). Ramp and value:

$$\text{ramp}(t) = 1 - \exp\left(-\frac{1}{2}\left(\frac{t}{t_{\text{ramp}}}\right)^2\right); \qquad \text{source_val} = \text{ramp}(t) \cdot \sin(2\pi f_0\, t\, \Delta t + \phi). \tag{17}$$

Default $t_{\text{ramp}} = 30$ time steps; $\phi$ is the phase offset (e.g. for 4-quadrant optimization).

# 8 Summary Table

| Label | Quantity | Code location (approx.) |
|---|---|---|
| (1) | Courant time step | 2618–2620, 185–188 |
| (2) | Actual $\Delta t$ | 2621, 188 |
| (3) | Effective $\varepsilon$, cond | 2684–2690 |
| (4), (5) | D-field update | 384–462 |
| (6) | E from D (ADE) | 508–513 |
| (7), (8) | H-field update | 544–619 |
| (9) | Instantaneous SAR | 762–784 |
| (10) | RMS SAR | 803–826 |
| (11) | SAR from phasor | 1077–1085 |
| (12) | Heat source $Q$ | 4358 |
| (13) | Pennes steady-state | 834–836 |
| (14) | Gauss-Seidel $T$ update | 853–861 |
| (15) | J-ratio | 1089–1098 |
| (16) | Penalized objective | 1102–1122 |
| (17) | CW ramp and source | 1005–1008, 3622–3625 |